

Zinc-Mediated Synthesis of α -C-Glycosides from 1,2-Anhydroglycosides

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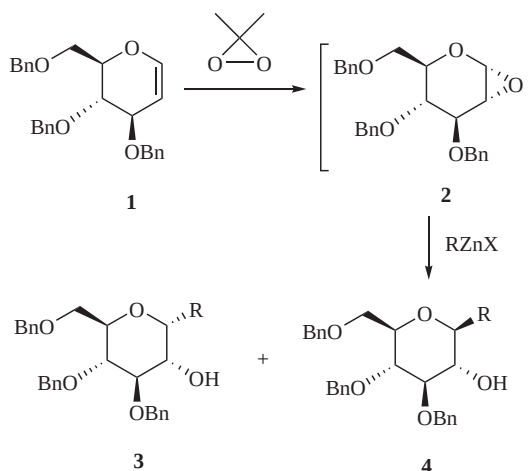
Abstract: α -C-Glycosides have been prepared by the addition of organozinc reagents to glycal epoxides. Dialkyl and diaryl zinc reagents in the presence of CF_3COOH provide the corresponding α -glycosides in 53–82% yields. Organozinc chlorides RZnCl formed by the reaction RLi with zinc chloride can also be successfully applied to this coupling reaction with high α -selectivity.

Key words: α -C-glycosides, epoxide, organozinc reagents, stereo-selectivity, trifluoroacetic acid

C-Glycosides are conformationally stable glycoside analogues, which have received a great deal of attention from the synthesis and medicinal chemistry.¹ Due to the importance of the stereochemical outcome of the C-glycosides, which must be at least highly stereoselective, if not stereospecific, great effort has been made to develop a number of elegant approaches to their synthesis.² The addition of organometallic reagents to glycal epoxides is a fundamental preparation of these compounds. Grignard reagents,³ copper reagents,⁴ and organostannanes^{5,4b} generally prefer to give β -C-glycosides, which have a *trans* relationship between the C-2 hydroxy group and the formed C-1 carbon-carbon bond. While the addition of aluminum and boron reagents⁶ to glycal epoxides has been reported to provide α -C-glycosides with good yields. Another synthetic approach to α -C-glycosides from epoxides concerns the reductive ring opening of epoxides with titanocene(III) chloride in the presence of trapping reagents.⁷

Here, we wish to report our preliminary results regarding the addition of organozinc species to glycal epoxides (Scheme 1). The *cis* addition products previously obtained in the addition of organozinc reagents to cyclo-1,3-diene monoepoxide prompted us to verify the possibility of zinc-mediated synthesis of α -C-Glycosides from 1,2-anhydroglycosides.⁸

The epoxidation of tribenzyl-D-glucal **1** according to Danishefsky's protocol⁹ followed by the addition of $\text{CF}_3\text{CO}_2\text{ZnEt}$, which is in situ formed by the reaction of ZnEt_2 with CF_3COOH in CH_2Cl_2 , gives α -C-glycoside **3a** with complete stereoselectivity in a 65% yield (entry 1 in Table 1).¹⁰ The stereochemical elucidation of the addition product is based on the analysis of ^1H NMR spectra. Indeed, the ^1H NMR spectrum of the corresponding O-2-acetylated derivative of **3a** shows a large coupling con-



Scheme 1

stant ($J_{2,3} = 8.8$ Hz) and a smaller coupling constant ($J_{1,2} = 5.5$ Hz) at $\delta = 5.05$ ppm. These data are consistent with prediction for α -C-glycoside.¹¹ The choice of Et_2O as solvent affords a 45% yield of **3a** after stirring for 16 hours. Reaction performed in THF provides only small amount of **3a** after lengthy reaction time (24 h).

The reaction is applicable to a variety of zinc reagents.¹² Dimethylzinc in the presence of CF_3COOH behaves identically to give α -C-glycoside **3b** in 81% yield. Aryl zinc compounds, such as diphenylzinc and difurylzinc, can

Table 1 Addition of ZnR_2 to Epoxide **2** in the Presence of $\text{CF}_3\text{CO}_2\text{H}$

Entry	ZnR_2	Solvent	Time (h)	Product	Yield (%) ^a
1	ZnEt_2	CH_2Cl_2	3	3a	65
2	ZnEt_2	Et_2O	16	3a	45
3	ZnMe_2	CH_2Cl_2	4	3b	81
4	ZnPh_2	CH_2Cl_2	5	3c	64
5	$\text{Zn}(\text{furyl})_2$	CH_2Cl_2	5	3d	82
6	$\text{Zn}(\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl})_2$	CH_2Cl_2	3	3e	58
7	$\text{Zn}(\text{CH}_2\text{CH}_2\text{CH}_2\text{OAc})_2$	CH_2Cl_2	5	3f	53

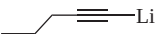
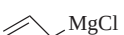
^a Isolated yields.

also be added to the epoxide **2** and afford the corresponding addition products in 64% and 82% yields respectively.

Importantly, functionalized zinc reagents proceed smoothly, and the reaction is able to withstand the inclusion of halide and ester (entries 6 and 7 in Table 1).

Encouraged by this initial result, we wish to extend further the organozinc reagents ZnR_2 to zinc halide RZnX , since it is atom efficient. Treatment of *n*-BuZnCl, which is easily prepared by the reaction of *n*-butyllithium with anhydrous zinc chloride in diethyl ether, with the glycal epoxide **2** affords the corresponding addition product with high α selectivity in 69% yield (entry 2 in Table 2). Use of higher concentration of organozinc specie *n*-BuZnCl could improve the reaction efficiency (entry 1 vs. entry 2 in Table 2). According to the conditions optimized for organozinc halide addition reaction, phenyl and furyl lithium in this manner also give the desired α -C-glycoside in 78% and 72% yields, respectively. Similarly, the reaction of acetylenic lithium with ZnCl_2 and epoxide **2** give only the products of α -C-glycosides in good yields.^{13,14}

Table 2 Addition of RLi or Grignard Reagents to Epoxide **2** in Presence of ZnCl_2^a

Entry	RM	Time (h)	Product	Yield (%) ^b	α/β^c
1 ^d	<i>n</i> -BuLi	16	3g	41	>95/5
2	<i>n</i> -BuLi	5	3g	69	>95/5
3	PhLi	5	3c	78	>95/5
4		5	3d	72	>95/5
5	Ph-C $\equiv$ C-Li	5	3h	86	100/0
6		5	3i	80	100/0
7	<i>n</i> -BuMgCl	4	3g	41	66/34
8		4	4a	74	0/100

^a Reaction condition: epoxide is added to 4.0 equiv of preformed reagent (RM, ZnCl_2), CH_2Cl_2 , 0 °C to r.t.

^b Isolated yields.

^c Determined by ^1H NMR.

^d 2.0 Equiv of reagent is used.

Of note are attempted coupling reactions with Grignard reagents in the presence of anhydrous zinc chloride. Treatment of *n*-butylmagnesium chloride with ZnCl_2 , followed by epoxide **2**, provides a 66:34 mixture of α - and β -C-glycosides (entry 7 in Table 2).¹⁵ On the other hand, the reaction of allylmagnesium chloride with ZnCl_2 and epoxide **2** affords exclusively the β -C-glycoside (entry 8 in Table 2). Essentially the same result was obtained when the above reaction was carried out in the absence of anhydrous zinc chloride.³ We intend to further study these reactions to improve both yield and stereochemical control.

The mechanism is closely related to that proposed by J. D. Rainier in his work.^{6b} The transfer of R group from RZnX , which can act as a Lewis acid system as well as being nucleophilic in character, occurs from the same face as the oxygen of epoxide and results in a *cis* addition product.

In conclusion, we have demonstrated a new method for the synthesis of α -C-glycosides using organozinc species. $\text{CF}_3\text{CO}_2\text{ZnR}$ affords α -glycosides with completely stereocontrol, whereas organozinc chloride RZnCl provides addition products with different degree of stereoselectivity, depending upon the organometallic reagent used. Additional studies of the scope are in progress.

Acknowledgment

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- General Procedure for Preparation of α -C-Glycosides.** To a solution of tri-O-benzyl-D-glucal (42 mg, 0.1 mmol) and 1.5 mL of CH_2Cl_2 at 0 °C was added a solution of dimethyldioxirane (0.15 mmol) in actone dropwise. After stirring for 30 min, the reaction mixture was concentrated in vacuum. The resulting white solid was taken up in CH_2Cl_2 (1.0 mL). To another solution of ZnR_2 (0.2 mmol, 2 equiv) in 1 mL of CH_2Cl_2 at 0 °C was added CF_3COOH (15 μL , 0.2 mmol, 2 equiv) very slowly via syringe under N_2 . After stirring for 30 min, to this solution was added the solution of epoxide dropwise via syringe. The mixture was stirred for 3–5 h from 0 °C to r.t. The reaction was quenched with sat. aq NH_4Cl and extracted with CH_2Cl_2 (3 $\times$ 10 mL). The

combined organic solution was washed with brine, dried (MgSO_4), and concentrated. Then, the product was isolated by silica gel column chromatography. Acetylated **3a**: Mp 55–56 °C. ^1H NMR (400 MHz, CDCl_3): δ = 7.33–7.27 (m, 13 H), 7.16 (m, 2 H), 5.05 (dd, J = 8.8, 5.5 Hz, 1 H), 4.77 (d, J = 11.5 Hz, 1 H), 4.74 (d, J = 9.5 Hz, 1 H), 4.71 (d, J = 11.2 Hz, 1 H), 4.63 (d, J = 12.2 Hz, 1 H), 4.52 (d, J = 10.3 Hz, 1 H), 4.50 (d, J = 10.9 Hz, 1 H), 4.02 (m, 1 H), 3.82 (m, 1 H), 3.71–3.65 (m, 4 H), 1.99 (s, 3 H), 1.70 (m, 1 H), 1.48 (m, 1 H), 0.94 (t, J = 7.4 Hz, 3 H). ^{13}C NMR (400 MHz, CDCl_3): δ = 170.1, 138.4, 138.2, 138.0, 128.4, 128.3, 128.0, 127.8, 127.7, 127.6, 80.2, 77.7, 74.7, 74.6, 73.7, 73.5, 73.1, 71.6, 69.0, 20.9, 19.1, 9.6. HRMS (FAB⁺) calcd for $\text{C}_{31}\text{H}_{36}\text{O}_6$: 505.2590 [MH^+]. Found: 505.2585. Acetylated **3e**: Mp 62–63 °C. ^1H NMR (400 MHz, CDCl_3): δ = 7.33–7.26 (m, 13 H), 7.15 (m, 2 H), 5.05 (dd, J = 8.6, 5.3 Hz, 1 H), 4.76 (d, J = 11.5 Hz, 1 H), 4.74 (d, J = 10.4 Hz, 1 H), 4.71 (d, J = 9.8 Hz, 1 H), 4.61 (d, J = 12.1 Hz, 1 H), 4.52 (d, J = 12.0 Hz, 1 H), 4.49 (d, J = 10.8 Hz, 1 H), 4.11 (m, 1 H), 3.82 (m, 1 H), 3.72–3.65 (m, 4 H), 3.51 (t, J = 6.6 Hz, 2 H), 1.99 (s, 3 H), 1.80–1.77 (m, 3 H), 1.55 (m, 1 H), 1.43–1.40 (m, 2 H). ^{13}C NMR (400 MHz, CDCl_3): δ = 170.1, 138.4, 138.1, 137.9, 129.7, 128.4, 128.3, 127.9, 127.8, 127.7, 127.6, 80.0, 77.5, 74.7, 74.6, 73.5, 72.9, 71.9, 68.9, 44.8, 32.2, 25.4, 22.6, 20.9. HRMS (FAB⁺) calcd for $\text{C}_{33}\text{H}_{39}\text{O}_6\text{Cl}$: 567.2513 [MH^+]. Found: 567.2500. Acetylated **3f**: Mp 77–78 °C. ^1H NMR (400 MHz, CDCl_3): δ = 7.27–7.10 (m, 13 H), 7.10 (m, 2 H), 4.98 (dd, J = 8.7, 5.4 Hz, 1 H), 4.71 (d, J = 11.8 Hz, 1 H), 4.67 (d, J = 10.7 Hz, 1 H), 4.65 (d, J = 11.1 Hz, 1 H), 4.55 (d, J = 12.1 Hz, 1 H), 4.46 (d, J = 11.9 Hz, 1 H), 4.44 (d, J = 10.9 Hz, 1 H), 4.09–3.94 (m, 3 H), 3.74 (m, 1 H), 3.66–3.59 (m, 4 H), 1.96 (s, 3 H), 1.93 (s, 3 H), 1.71–1.51 (m, 4 H), 1.47–1.26 (m, 2 H). ^{13}C NMR (400 MHz, CDCl_3): δ = 171.2, 170.1, 138.4, 138.2, 138.1, 128.5, 128.4, 128.1,

127.9, 127.8, 127.7, 127.6, 80.1, 77.6, 74.8, 74.7, 73.6, 73.1, 72.1, 72.0, 69.0, 64.3, 29.8, 28.4, 25.7, 21.8, 21.0. HRMS (FAB⁺) calcd for $\text{C}_{35}\text{H}_{42}\text{O}_8$: 591.2958 [MH^+]. Found: 591.2952. Acetylated **3h**: Mp 70–71 °C. ^1H NMR (300 MHz, CDCl_3): δ = 7.38 (m, 2 H), 7.29–7.19 (m, 16 H), 7.08 (m, 2 H), 5.13 (d, J = 5.7 Hz, 1 H), 4.91 (dd, J = 9.9, 5.7 Hz, 1 H), 4.81–4.71 (m, 3 H), 4.58 (d, J = 12.0 Hz, 1 H), 4.48 (d, J = 10.8 Hz, 1 H), 4.46 (d, J = 12.3 Hz, 1 H), 3.98 (m, 2 H), 3.71–3.63 (m, 3 H), 1.96 (s, 3 H). ^{13}C NMR (300 MHz, CDCl_3): δ = 170.2, 138.7, 138.2, 138.1, 132.1, 128.6, 128.5, 128.2, 128.1, 127.9, 127.8, 122.3, 89.6, 83.2, 81.4, 77.9, 75.5, 75.4, 74.2, 73.7, 72.8, 68.8, 66.3, 21.1. IR (KBr): ν = 2909, 2228, 1741, 1454, 1238, 1099, 1041 cm^{-1} . HRMS (FAB⁺) calcd for $\text{C}_{37}\text{H}_{36}\text{O}_6$: 599.2409 [MNa^+]. Found: 599.2408.

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- (15) For comparison, the reaction of *n*-butylmagnesium chloride with epoxide **2** in THF from –78 °C to 0 °C gave a 45:55 mixture of α - and β -C-glycoside.